

Relevance and surprise scoring smool.

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Objective

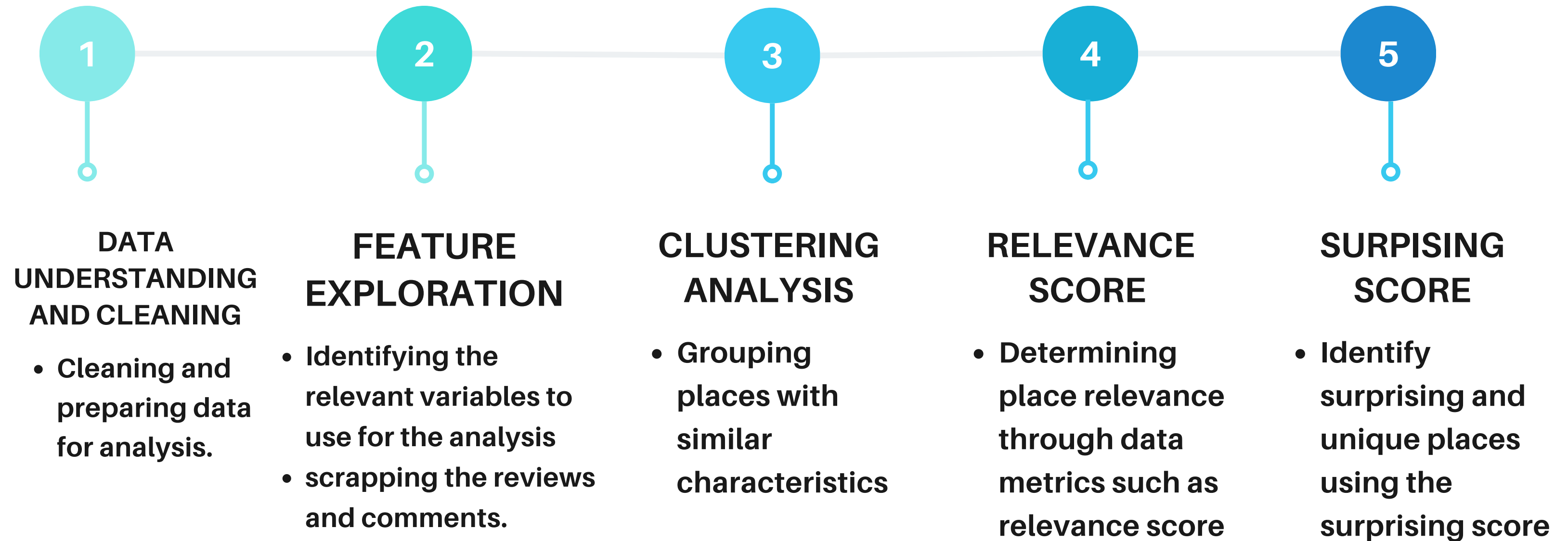
Identify the most interesting and special places using relevance and surprise scores.

Scope

Analyze data, establish scores, and leverage ML and DL algorithms.

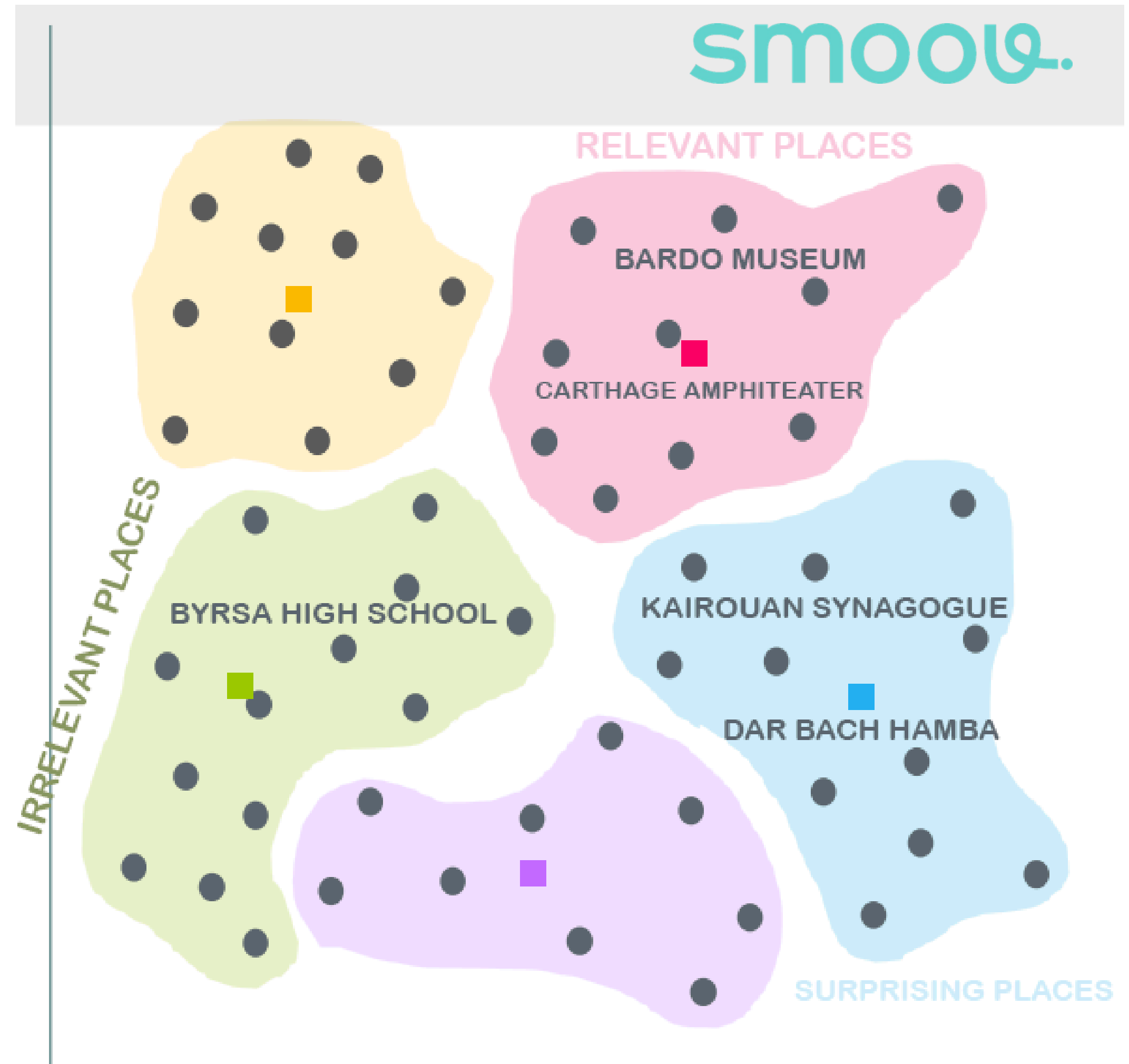


METHODOLOGY FOR IMPLEMENTING THE SCORES



Clustering Analysis

- Grouping places with similar characteristics using algorithms such as KMeans and self organizing maps
- Label clusters based on their dominant features (e.g., highly relevant, highly surprising,)



Relevance Score Calculation

Wikipedia Data:

- Fetch content using APIs.
- Calculate word count for relevance.

Review Extraction and Sentiment

Analysis:

- Collect reviews using APIs/web scraping.
- Metrics: Average sentiment score, number of reviews, sentiment polarity.

UNESCO and Heritage Sites:

- Assign higher weights to UNESCO/heritage sites.

✓ Pros

 1100	 18	Rich history beautifully preserved
 993	 24	Wealth of educational knowledge
 835	 86	Highly educational and well-curated
 737	 39	Immersive authentic experience
 723	 33	Genuine cultural reflection
 709	 27	Suitable for a wide range of ages

✗ Cons

 642	 307	Overcrowded and noisy
 504	 232	Poorly maintained facilities
 516	 294	Poorly maintained facilities
 330	 344	Some people may find it forgettable or boring
 207	 366	"Limited accessibility options."

Surprise Score Calculation

Historical and Cultural Significance:

- use Text mining for cultural/historical significance by interpreting the reviews of the visitors.

Unique Features:

- Quantify uniqueness (architecture, rare species).

Visitor Trends:

- Analyze visit patterns for unexpected places.

Integration:

- Combine significance, uniqueness, and trends into a surprise score.



Relevant Features

- **Wikipedia Word Count (WWC):** Number of words on the Wikipedia page for the place.
- **Review Sentiment Score (RSS):** Average sentiment score of reviews from sources like Google Maps.
- **Number of Reviews (NR):** Total number of reviews.
- **UNESCO Status (UNESCO):** Binary variable (1 if UNESCO World Heritage site, 0 otherwise).
- **Heritage Status (Heritage):** Binary variable (1 if heritage site, 0 otherwise).
- **Category (Cat):** Categorical variable ranging from 1 to 6.
- **Type of Place (Type):** Encoded type of place (e.g., 1 for historical, 2 for natural, etc.).

Weighted Formula for Relevance Score

$$R = w_1 \cdot \text{WWC} + w_2 \cdot \text{RSS} + w_3 \cdot \log(1 + \text{NR}) + w_4 \cdot \text{UNESCO} + w_5 \cdot \text{Herita} \\ w_6 \cdot \text{Cat} + w_7 \cdot \text{Type}$$

Summary

We implemented a comprehensive approach to deducing relevance and surprise scores

Conclusion

The resulting scores help identify the most relevant and special places. By displaying these places in the Smoov application, users can sort them by relevance score and surprise score, making it easier to discover interesting locations.



smoov.